

GAEA-Net: Generating Activity-Enriched Abnormal ECGs via Adversarial Network

APPENDIX

Appendix A: Datasets

1.PXB-XL

PTB-XL consists of 21,837 12-lead ECG records from 18,885 patients. The dataset includes covers 71 diagnostic, form, and rhythm statements assigned by up to two cardiologists. It also contains rich demographic metadata such as age, gender, and infarction characteristics across diverse populations and clinical conditions.

2.PTB

PTB contains 549 fifteen-lead ECG recordings (12 standard leads) acquired at 1 kHz, 16-bit resolution (± 16.384 mV) from 290 subjects aged 17–87 yr (209 males, 81 females).

3.MuseECG

MuseECG contains 10,646 12-lead resting ECG recordings, each 10 seconds in duration and sampled at 500Hz. All recordings were collected using the GE MUSE system and annotated by two certified cardiologists, with final arbitration by a senior physician in case of disagreement. Each sample includes both rhythm labels and auxiliary diagnostic conditions.

4.MedalCare-XL

MedalCare-XL contains 16,900 synthetic 12-lead ECG signals generated from mechanistic cardiac electrophysiology models, offering physiologically grounded and label-consistent simulations. The dataset includes balanced representations of healthy individuals and seven cardiac pathology classes.

5.CinC 2017

CinC 2017 dataset comprises 8,528 single-lead ECG recordings with durations ranging from 9 to 61 seconds, sampled at 300 Hz. The recordings were acquired using the AliveCor portable device and labeled into four rhythm categories: normal sinus rhythm, atrial fibrillation (AF), and other abnormal rhythms.

Appendix B: Baseline Implementation Details

1.ECG Twin

ECG Twin develops a vector-quantized feature separator that distinguishes between disease and normal segments of ECG signals under the guidance of diagnostic reports. To incorporate their methodology, we utilize their open-source code available at <https://github.com/huyjj/LAVQ-Editor>. Notably, since our datasets do not include pathology reports, we have omitted the text encoder component from the model architecture. During training, we set the number of epochs to 300, the learning rate to 0.001, and the batch size to 128 on all datasets.

2.PGAN

PGAN introduces LSTM into GANs to generate plausible ECG signals, where the generator consists of a BiLSTM with 128 units. Under the instructions provided in its GitHub at <https://github.com/MikhailMurashov/ecgGAN>, we have rewritten the code using PyTorch for our experiments. We set the number of epochs to 100, and the batch size to 128 on all datasets. To address the issue of an overly strong discriminator during training, we adjust the learning rate of the generator to 0.0002 and the learning rate of the discriminator to 0.00001, training the discriminator once every five batches.

3.CardioGAN

CardioGAN is an attentive generative adversarial network, featuring a U-Net generator with attention mechanisms and two specialized discriminators that evaluate both temporal and spectral signal characteristics. We utilize their open-source code available at <https://github.com/pritamqu/ppg2ecg-cardiogán>. Our implementation repurposes this architecture to translate between different ECG morphologies as categorized in the PTB-XL dataset, MuseECG Dataset and MedalCare-XL Dataset. Models were trained with a batch size of 128 for 40 epochs, using an initial learning rate of $1e-3$.

4.StyleGAN

StyleGAN3 is a state-of-the-art generative adversarial network (GAN) architecture primarily designed for generating high-quality images. We used the open-source code from the official repository: <https://github.com/NVlabs/stylegan3>. Although ECG signals are 1D, StyleGAN3 is typically applied to 2D images, so we adapted the approach by embedding the 1D ECG signals into 3-channel 2D images for training. Specifically, ECG time-series data are transformed into a 2D space using techniques such as GASF (Gramian Angular Summation Field), GADF (Gramian Angular Difference Field), and MTF (Markov Transition Field), which generate RGB-like 3-channel data. Finally, the original 1D ECG time-series are reconstructed using

the diagonals of the GASF channel. For training, the setup used 3 GPUs with a batch size of 36, 2000k images (kimg), and a batch per GPU of 12.

5. SimGAN

SimGAN integrates a biological simulator into a Generative Adversarial Network to synthesize ECG signals, building upon the open-source implementation available at https://github.com/DreamStudioAI/sim_gan.git. The framework incorporates domain-specific knowledge from a heart simulator, represented by a system of Ordinary Differential Equations (ODEs), to guide the generative process. To adapt the method to our dataset, we modified the original euler loss function, which was based on a single heartbeat cycle, by dividing the generated ECG signal into 7 segments and calculating the loss for each segment individually.

6. Diffusion-SSSM

Diffusion-SSSM integrates a diffusion model with a structured state space model (SSSM) to generate synthetic 12-lead electrocardiograms (ECGs) based on clinical data. This approach leverages the superior performance of diffusion models in data generation while incorporating the temporal dependencies inherent in ECG signals through the structured framework of the SSSM. To tailor the dataset to our specific experimental setup, we redefined the 71 original ECG statements as conditional categories. Additionally, we reduced the number of embedding dimensions accordingly to optimize model efficiency. The model was trained over 10,001 iterations, with a batch size of 4 and a learning rate set to $2e-4$.

Appendix C: ROC and AUC Results on Different Datasets

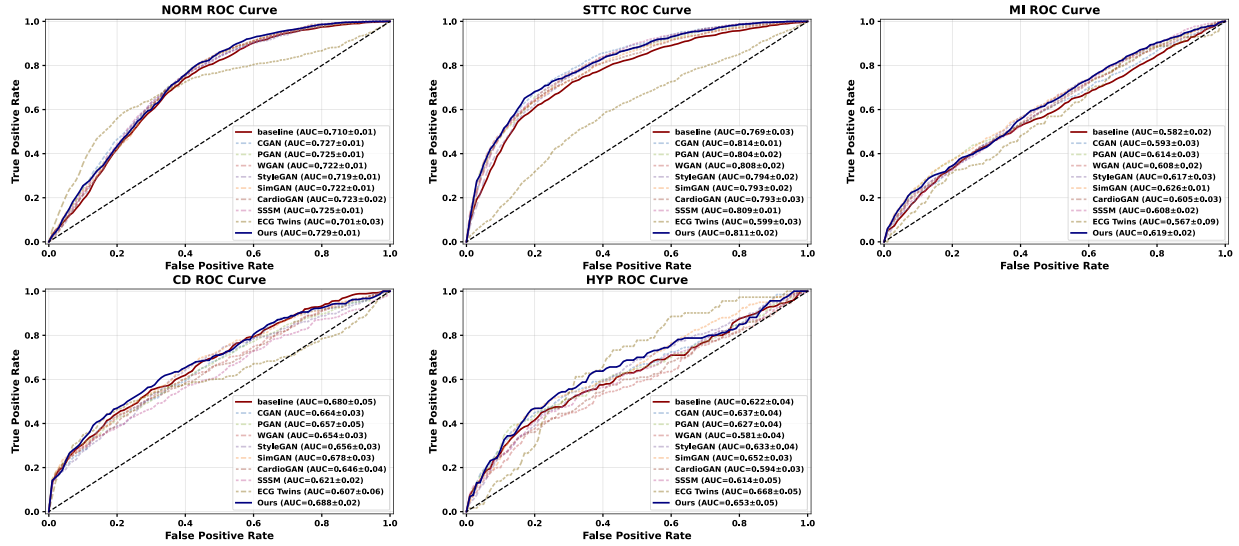


Fig. 1: Visualization of xxx

TABLE I: Class-wise AUC 95% confidence intervals of all methods on the PTB-XL dataset.

Method	Class1: NORM	Class2: STTC	Class3: MI	Class4: CD	Class5: HYP
Baseline	[0.696, 0.744]	[0.598, 0.613]	[0.517, 0.621]	[0.536, 0.577]	[0.586, 0.680]
CGAN	[0.736, 0.785]	[0.626, 0.648]	[0.559, 0.617]	[0.580, 0.646]	[0.558, 0.676]
PGAN	[0.705, 0.736]	[0.596, 0.618]	[0.543, 0.609]	[0.545, 0.585]	[0.593, 0.671]
WGAN	[0.714, 0.764]	[0.600, 0.651]	[0.572, 0.624]	[0.540, 0.590]	[0.609, 0.710]
StyleGAN	[0.752, 0.776]	[0.642, 0.658]	[0.516, 0.560]	[0.568, 0.601]	[0.563, 0.593]
SimGAN	[0.702, 0.764]	[0.579, 0.628]	[0.566, 0.657]	[0.537, 0.586]	[0.530, 0.604]
CardioGAN	[0.758, 0.787]	[0.630, 0.648]	[0.562, 0.619]	[0.529, 0.587]	[0.520, 0.657]
SSSM	[0.705, 0.748]	[0.609, 0.636]	[0.511, 0.561]	[0.583, 0.634]	[0.551, 0.660]
ECG Twins	[0.685, 0.718]	[0.583, 0.616]	[0.512, 0.622]	[0.573, 0.642]	[0.636, 0.699]
Ours	[0.759, 0.773]	[0.630, 0.641]	[0.534, 0.613]	[0.640, 0.674]	[0.668, 0.768]

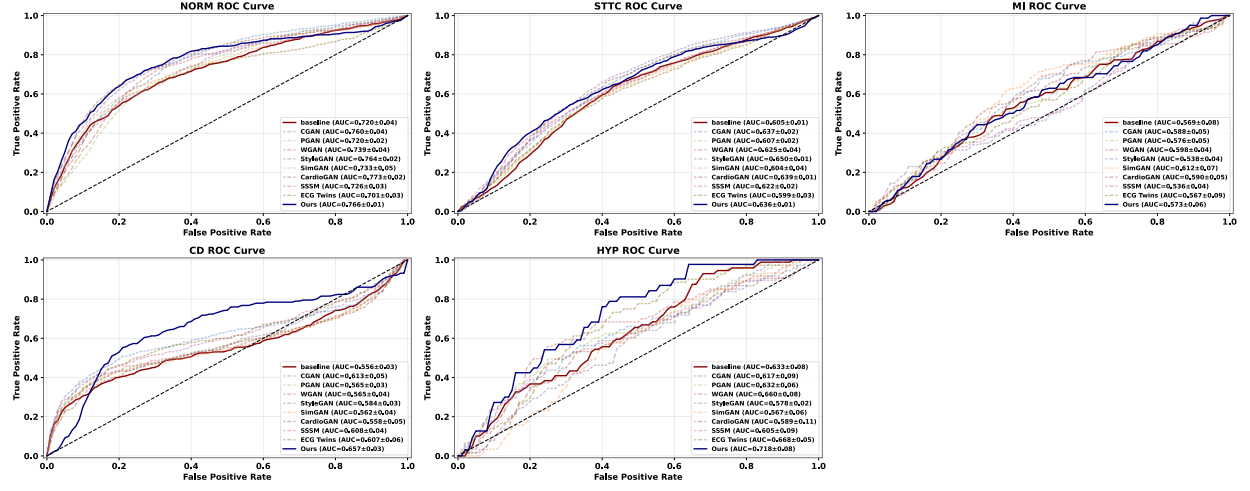


Fig. 2: Visualization of xxx

TABLE II: Class-wise AUC with 95% confidence intervals of all methods on the dataset.

Method	Class1: NORM	Class2: STTC	Class3: MI	Class4: CD	Class5: HYP
Baseline	[0.696, 0.744]	[0.598, 0.613]	[0.517, 0.621]	[0.536, 0.577]	[0.586, 0.680]
CGAN	[0.736, 0.785]	[0.626, 0.648]	[0.559, 0.617]	[0.580, 0.646]	[0.558, 0.676]
PGAN	[0.705, 0.736]	[0.596, 0.618]	[0.543, 0.609]	[0.545, 0.585]	[0.593, 0.671]
WGAN	[0.714, 0.764]	[0.600, 0.651]	[0.572, 0.624]	[0.540, 0.590]	[0.609, 0.710]
StyleGAN	[0.752, 0.776]	[0.642, 0.658]	[0.516, 0.560]	[0.568, 0.601]	[0.563, 0.593]
SimGAN	[0.702, 0.764]	[0.579, 0.628]	[0.566, 0.657]	[0.537, 0.561]	[0.530, 0.604]
CardioGAN	[0.758, 0.787]	[0.630, 0.648]	[0.562, 0.619]	[0.529, 0.587]	[0.520, 0.657]
SSSM	[0.705, 0.748]	[0.609, 0.636]	[0.511, 0.561]	[0.583, 0.634]	[0.551, 0.660]
ECG Twins	[0.685, 0.718]	[0.583, 0.616]	[0.512, 0.622]	[0.573, 0.642]	[0.636, 0.699]
Ours	[0.759, 0.773]	[0.631, 0.641]	[0.534, 0.613]	[0.640, 0.674]	[0.668, 0.768]

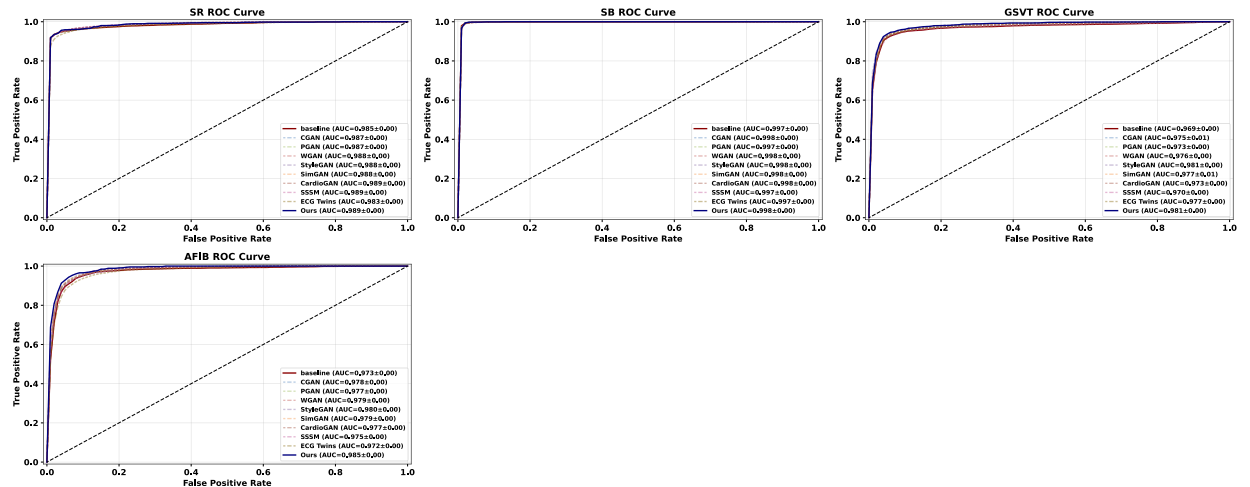


Fig. 3: Visualization of xxx

TABLE III: Class-wise AUC 95% confidence intervals of all methods on the PhysioNet dataset.

Method	Class1: SR	Class2: SB	Class3: GSVT	Class4: AFIB
Baseline	[0.984, 0.986]	[0.997, 0.997]	[0.966, 0.971]	[0.971, 0.974]
CGAN	[0.986, 0.988]	[0.997, 0.998]	[0.972, 0.978]	[0.976, 0.979]
PGAN	[0.986, 0.988]	[0.997, 0.998]	[0.971, 0.975]	[0.976, 0.978]
WGAN	[0.987, 0.989]	[0.997, 0.998]	[0.974, 0.979]	[0.978, 0.981]
StyleGAN	[0.987, 0.989]	[0.998, 0.998]	[0.979, 0.983]	[0.978, 0.982]
SimGAN	[0.986, 0.990]	[0.998, 0.998]	[0.973, 0.981]	[0.977, 0.982]
CardioGAN	[0.988, 0.990]	[0.997, 0.998]	[0.971, 0.976]	[0.976, 0.979]
SSSM	[0.988, 0.990]	[0.997, 0.997]	[0.968, 0.973]	[0.973, 0.977]
ECG Twins	[0.981, 0.985]	[0.996, 0.997]	[0.975, 0.980]	[0.970, 0.974]
Ours	[0.988, 0.990]	[0.998, 0.998]	[0.979, 0.984]	[0.985, 0.985]

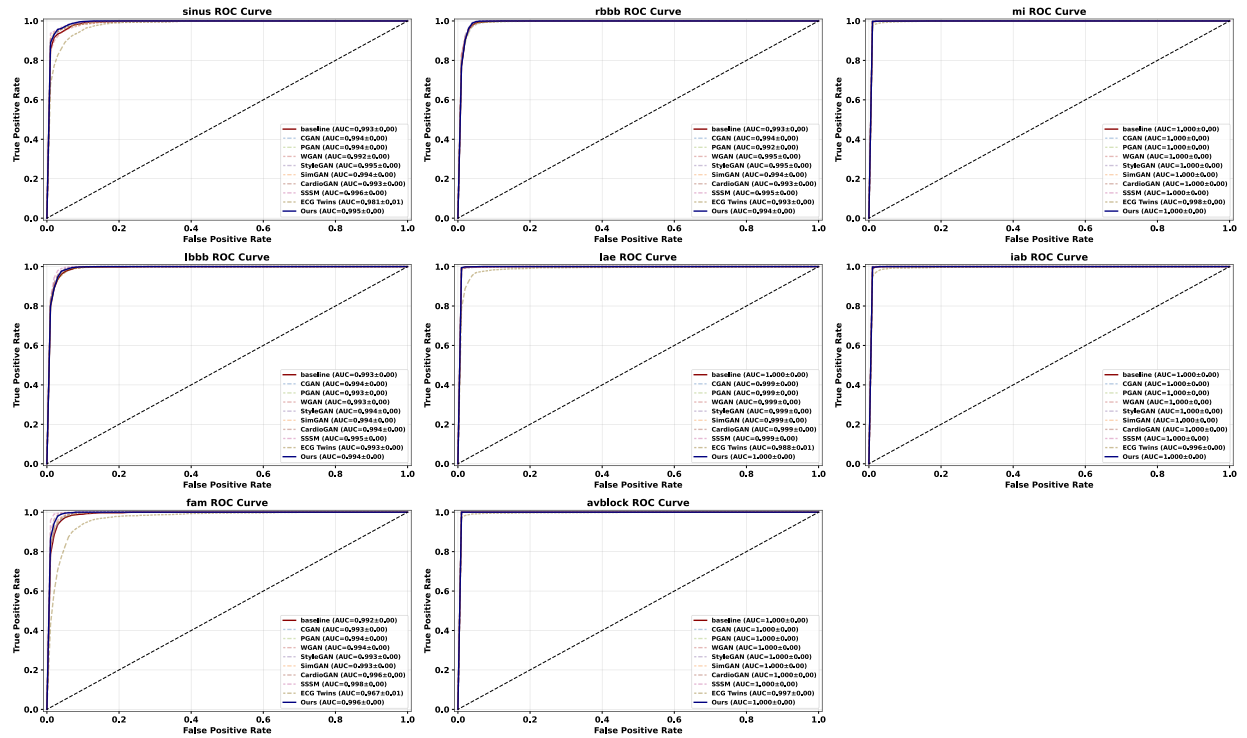


Fig. 4: Visualization of xxx

TABLE IV: Class-wise AUC with 95% confidence intervals of all methods on the dataset.

Method	sinus	rbbb	mi	lbbb	lac	iab	fam	avblock
Baseline	[0.991, 0.994]	[0.992, 0.994]	[0.997, 1.000]	[0.991, 0.994]	[1.000, 1.000]	[0.999, 1.000]	[0.990, 0.993]	[1.000, 1.000]
CGAN	[0.992, 0.996]	[0.993, 0.995]	[1.000, 1.000]	[0.993, 0.995]	[0.999, 1.000]	[1.000, 1.000]	[0.991, 0.996]	[1.000, 1.000]
PGAN	[0.993, 0.995]	[0.991, 0.994]	[1.000, 1.000]	[0.992, 0.994]	[0.999, 1.000]	[1.000, 1.000]	[0.991, 0.996]	[1.000, 1.000]
WGAN	[0.990, 0.994]	[0.994, 0.996]	[1.000, 1.000]	[0.991, 0.995]	[0.999, 1.000]	[1.000, 1.000]	[0.992, 0.996]	[1.000, 1.000]
StyleGAN	[0.994, 0.996]	[0.994, 0.995]	[1.000, 1.000]	[0.994, 0.995]	[0.999, 1.000]	[1.000, 1.000]	[0.991, 0.994]	[1.000, 1.000]
SimGAN	[0.993, 0.995]	[0.992, 0.995]	[1.000, 1.000]	[0.993, 0.994]	[0.999, 1.000]	[1.000, 1.000]	[0.992, 0.994]	[1.000, 1.000]
CardioGAN	[0.992, 0.994]	[0.992, 0.995]	[1.000, 1.000]	[0.993, 0.995]	[0.999, 1.000]	[1.000, 1.000]	[0.996, 0.997]	[1.000, 1.000]
SSSM	[0.995, 0.997]	[0.994, 0.995]	[1.000, 1.000]	[0.995, 0.996]	[0.999, 1.000]	[1.000, 1.000]	[0.998, 0.999]	[1.000, 1.000]
ECG Twins	[0.977, 0.985]	[0.992, 0.994]	[0.998, 0.999]	[0.993, 0.994]	[0.984, 0.992]	[0.994, 0.998]	[0.961, 0.974]	[0.996, 0.999]
Ours	[0.994, 0.996]	[0.993, 0.995]	[1.000, 1.000]	[0.993, 0.995]	[1.000, 1.000]	[1.000, 1.000]	[0.995, 0.997]	[1.000, 1.000]

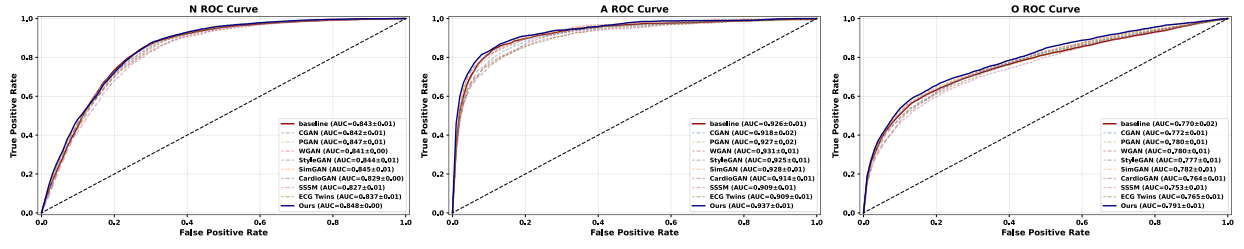


Fig. 5: Visualization of xxx

TABLE V: Class-wise AUC with 95% confidence intervals of all methods on the CPSC dataset.

Method	Class1: N	Class2: A	Class3: O
Baseline	[0.839, 0.846]	[0.920, 0.932]	[0.759, 0.781]
CGAN	[0.837, 0.847]	[0.908, 0.927]	[0.764, 0.779]
PGAN	[0.841, 0.853]	[0.917, 0.936]	[0.776, 0.785]
WGAN	[0.838, 0.844]	[0.928, 0.934]	[0.775, 0.785]
StyleGAN	[0.839, 0.849]	[0.920, 0.931]	[0.772, 0.783]
SimGAN	[0.841, 0.849]	[0.923, 0.933]	[0.776, 0.788]
CardioGAN	[0.827, 0.831]	[0.908, 0.920]	[0.760, 0.769]
SSSM	[0.821, 0.834]	[0.901, 0.916]	[0.747, 0.759]
ECG Twins	[0.832, 0.842]	[0.900, 0.918]	[0.759, 0.770]
Ours	[0.846, 0.851]	[0.934, 0.941]	[0.786, 0.796]

Appendix D: Disentanglement on Different Datasets

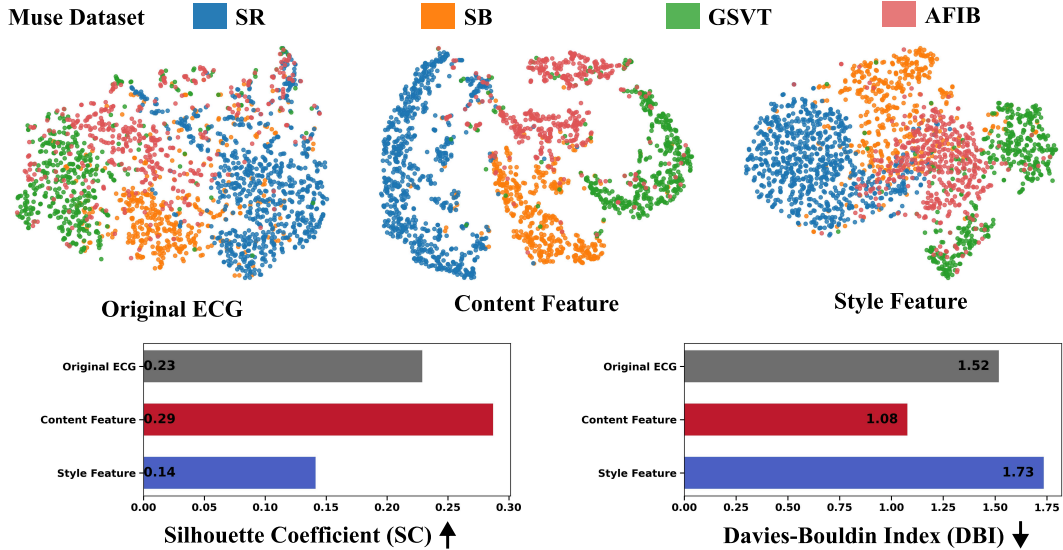


Fig. 6: Visualization of xxx

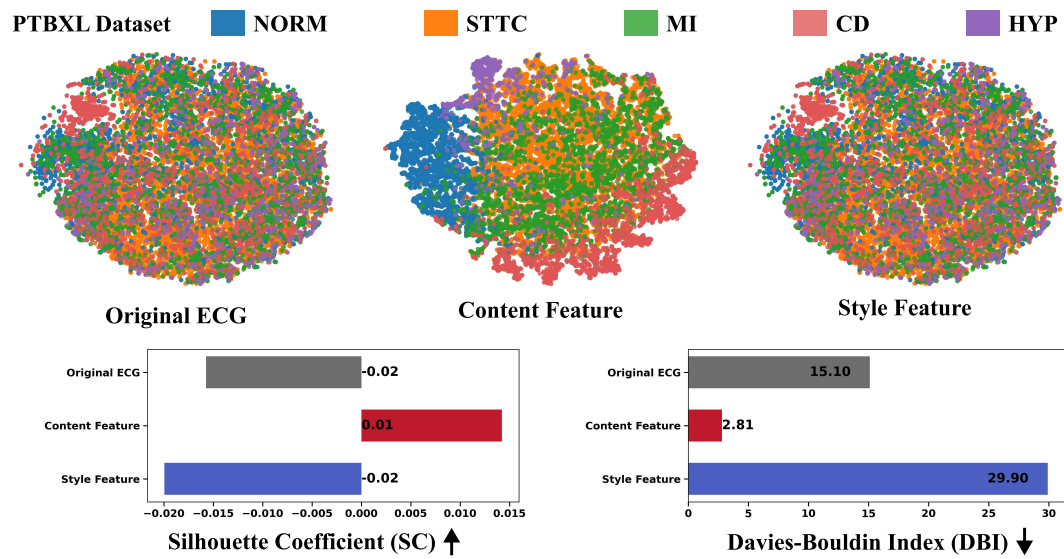


Fig. 7: Visualization of xxx